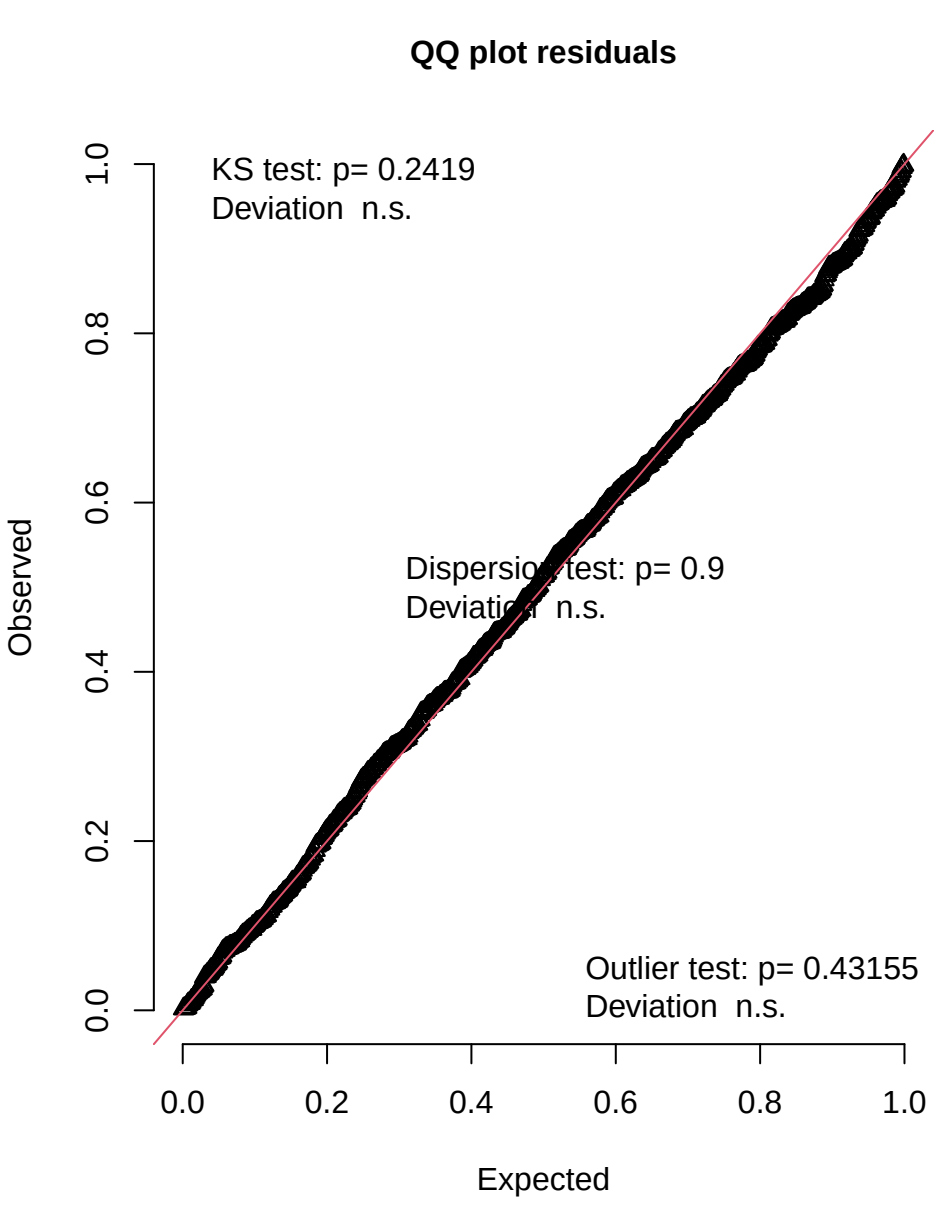
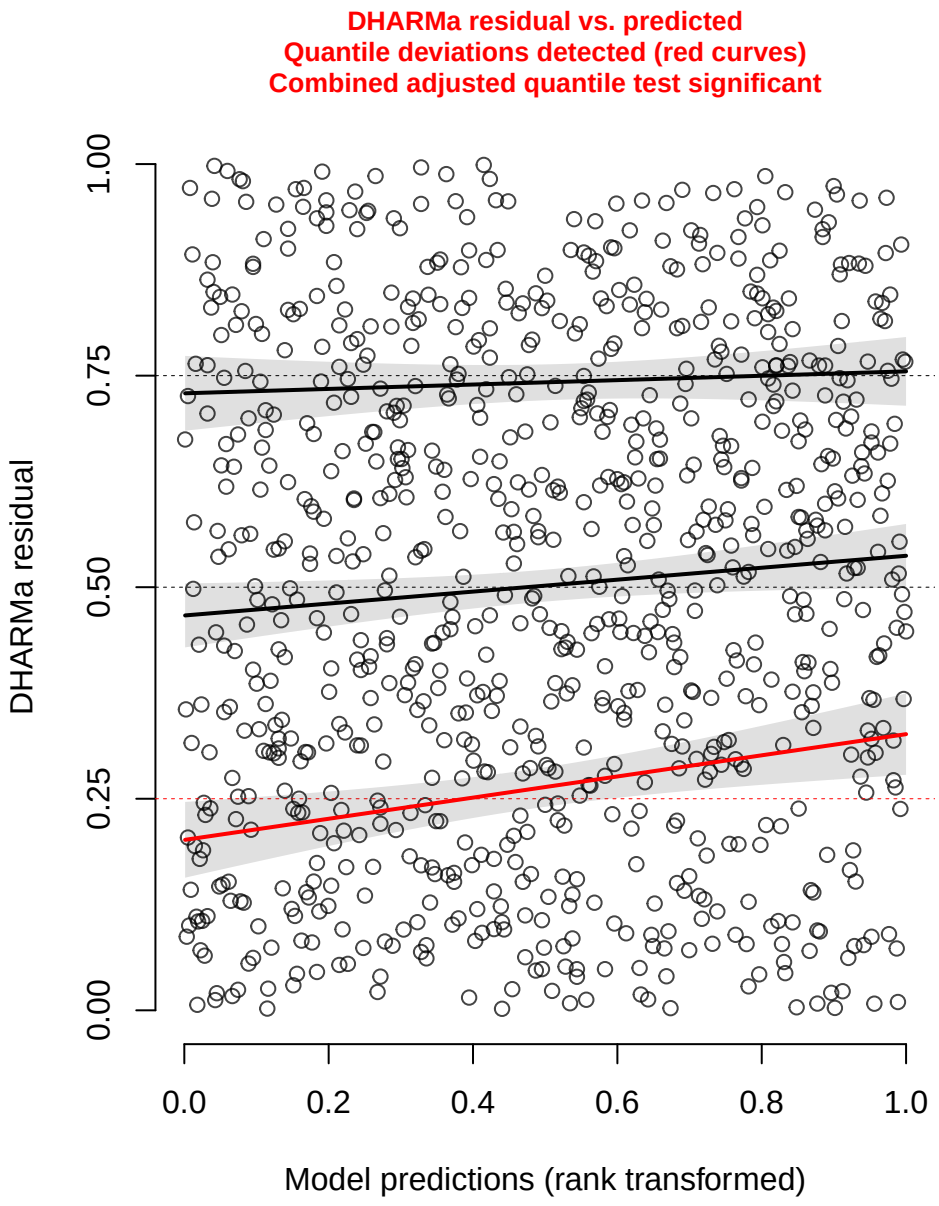
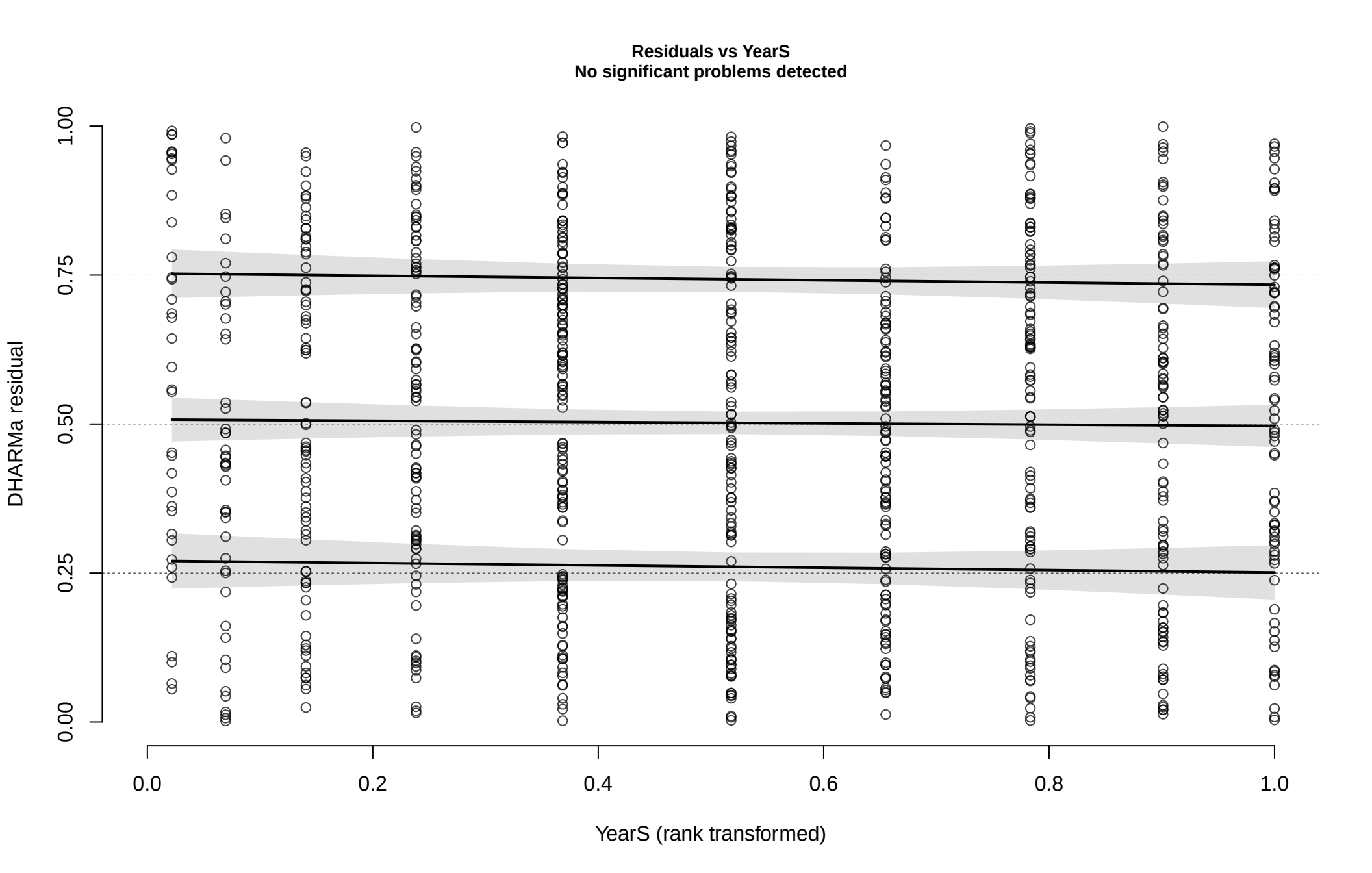
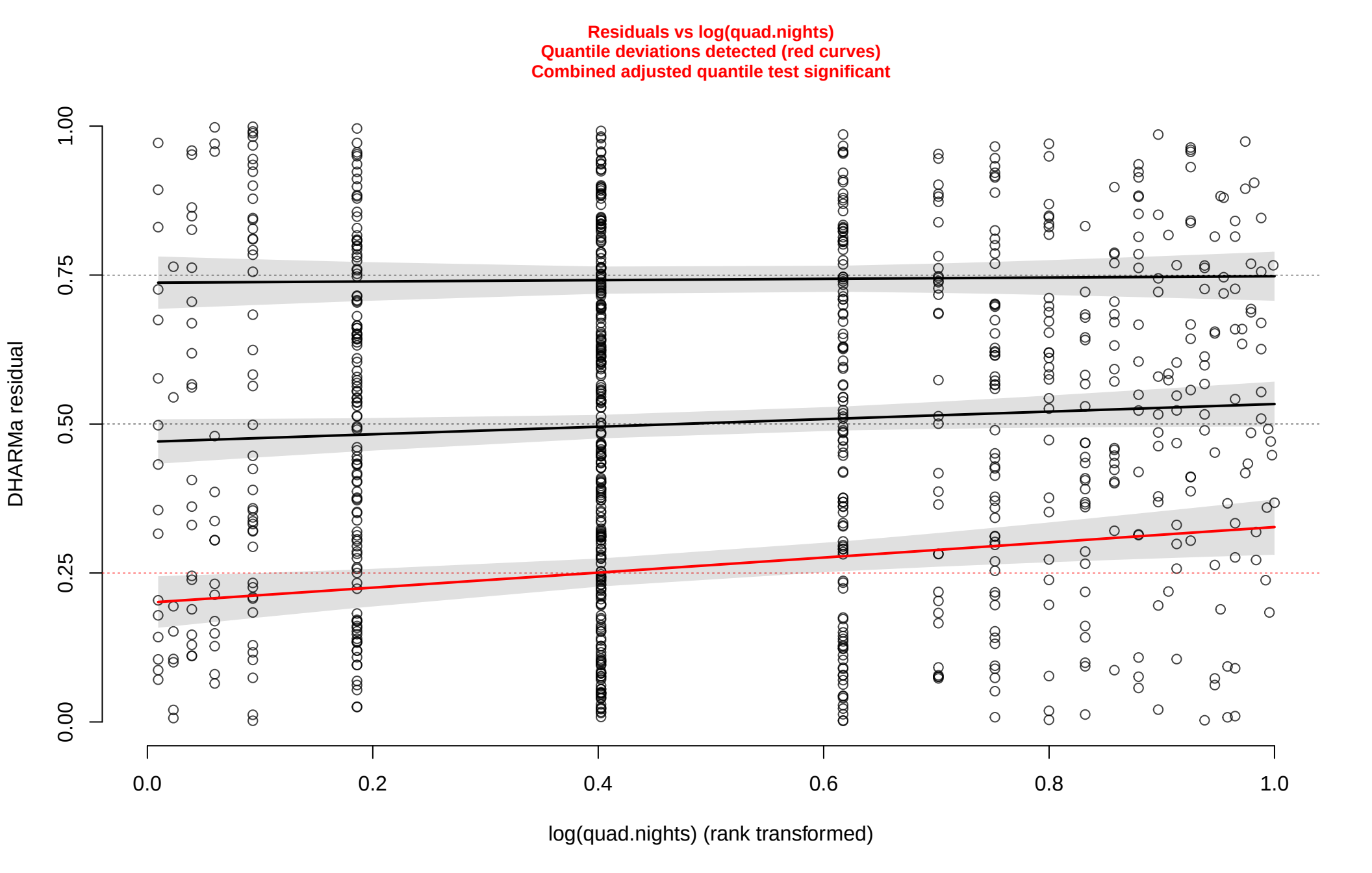
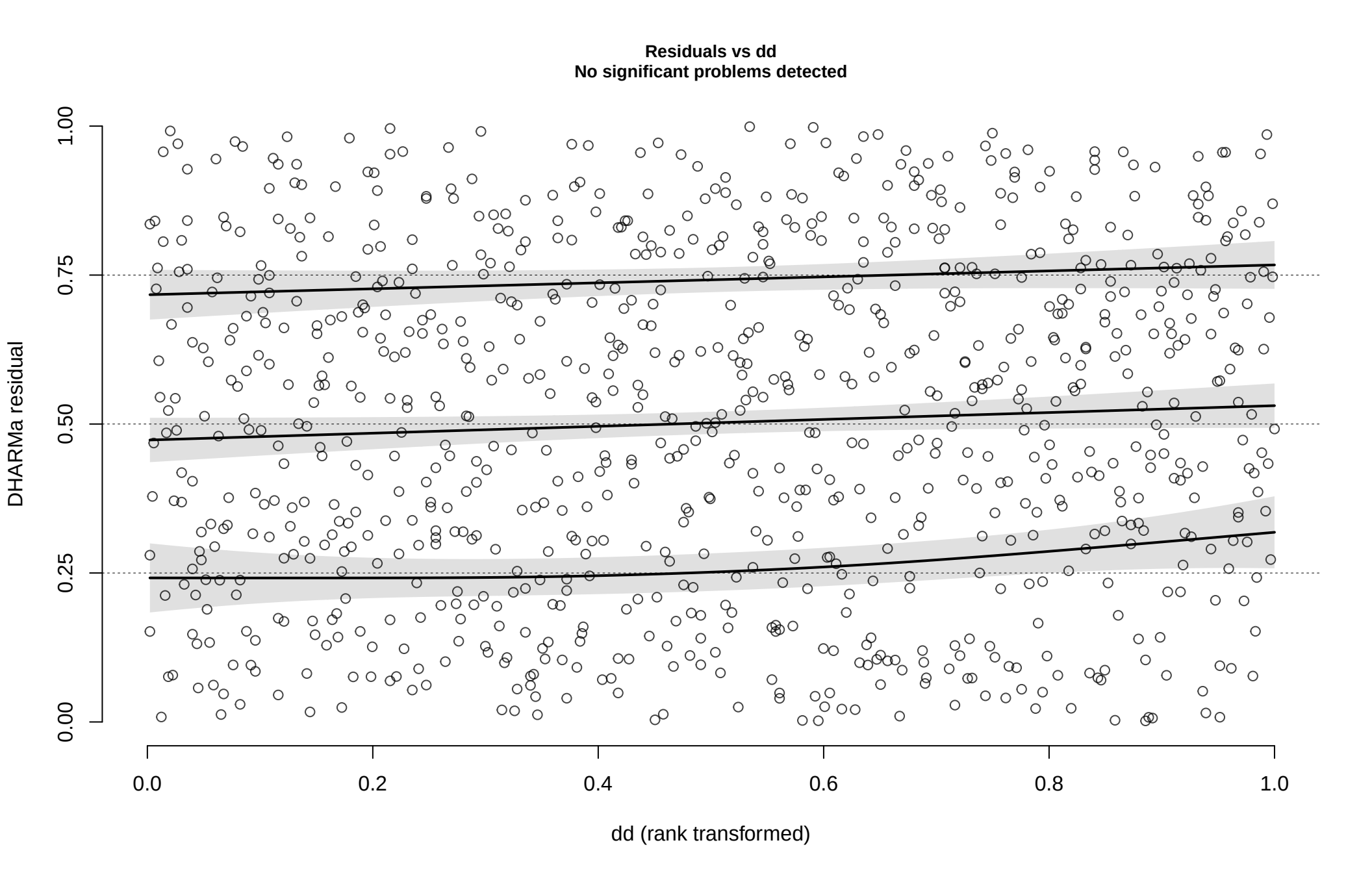
DHARMa residuals: COTO

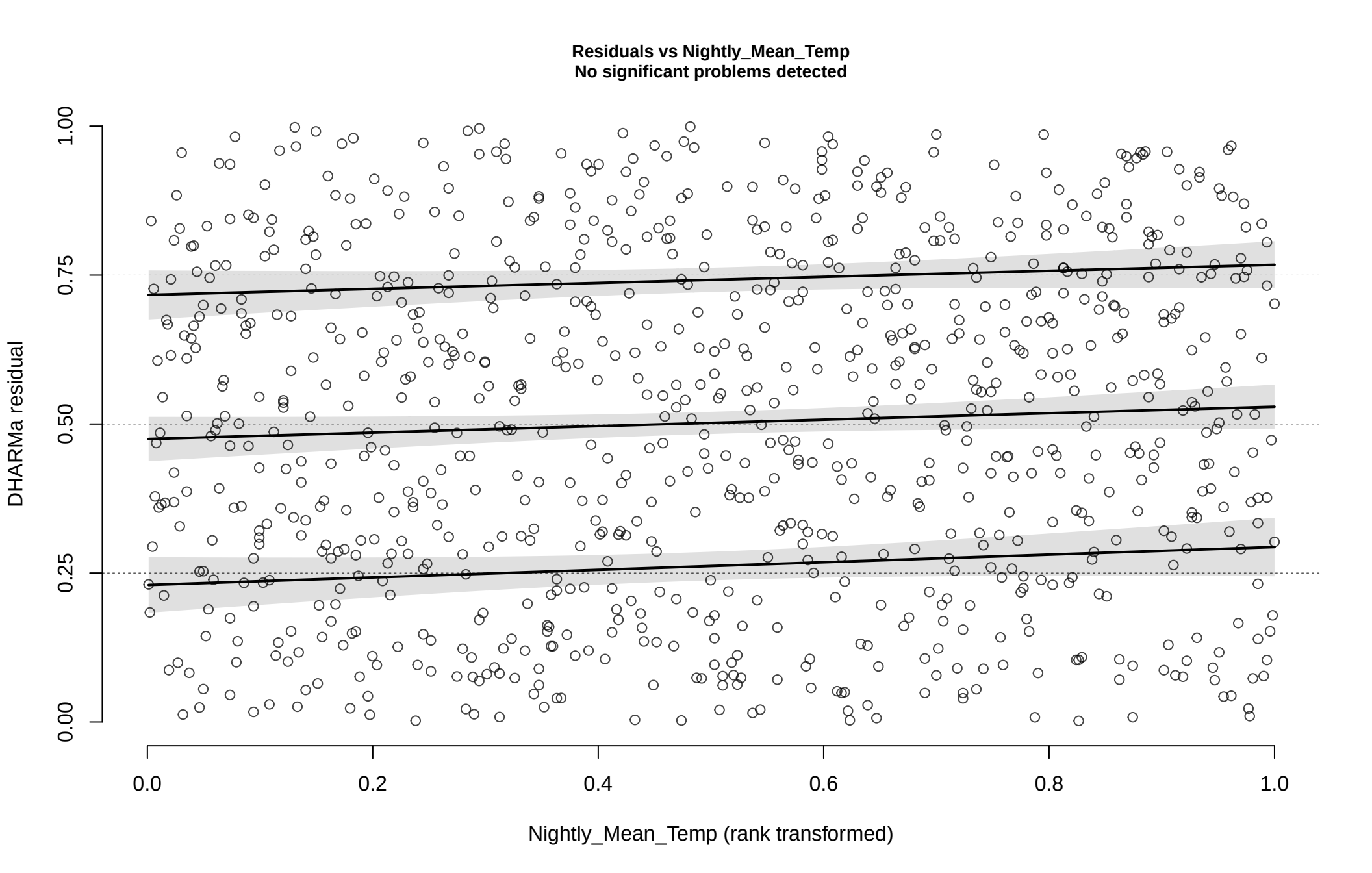


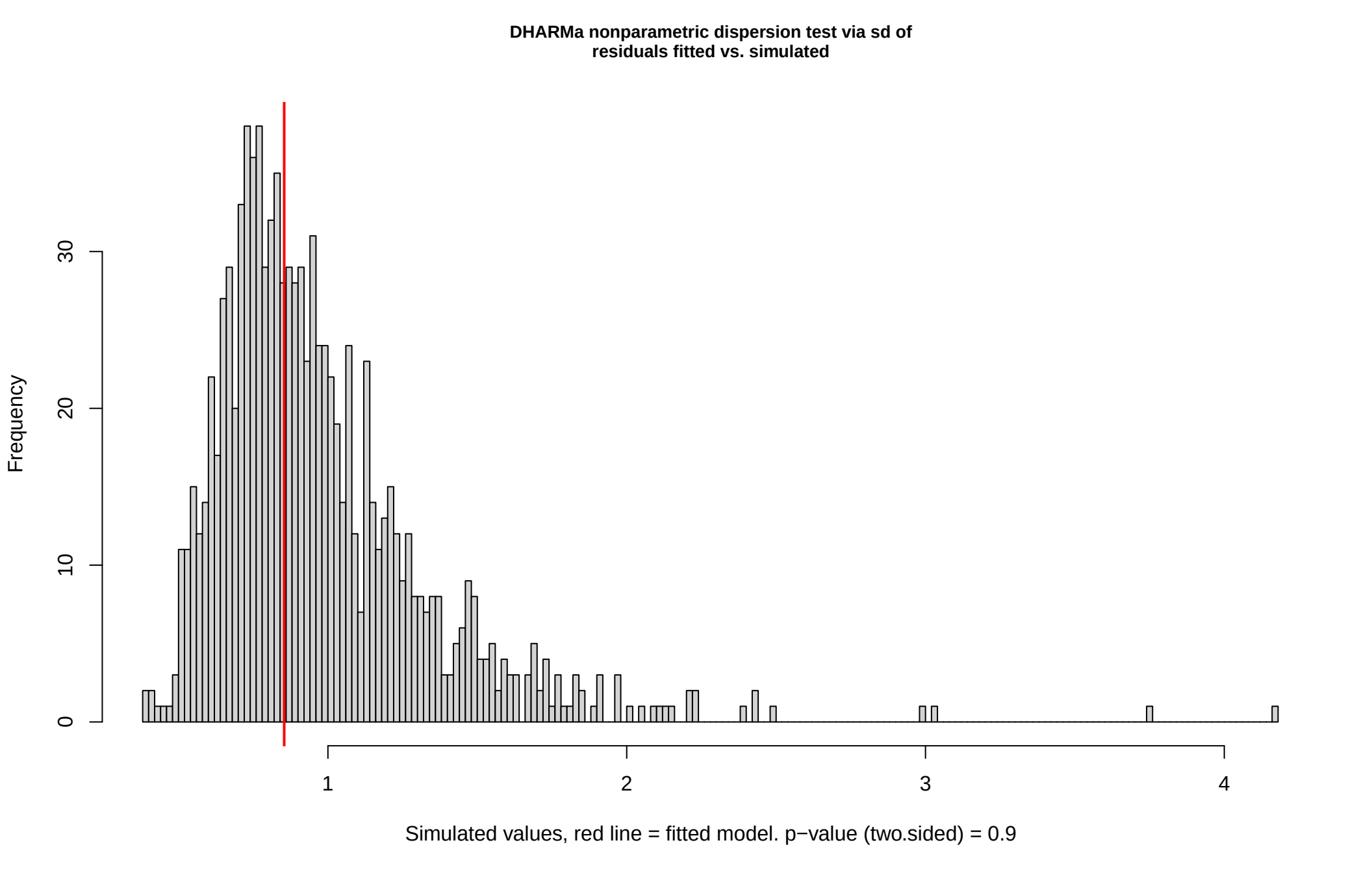



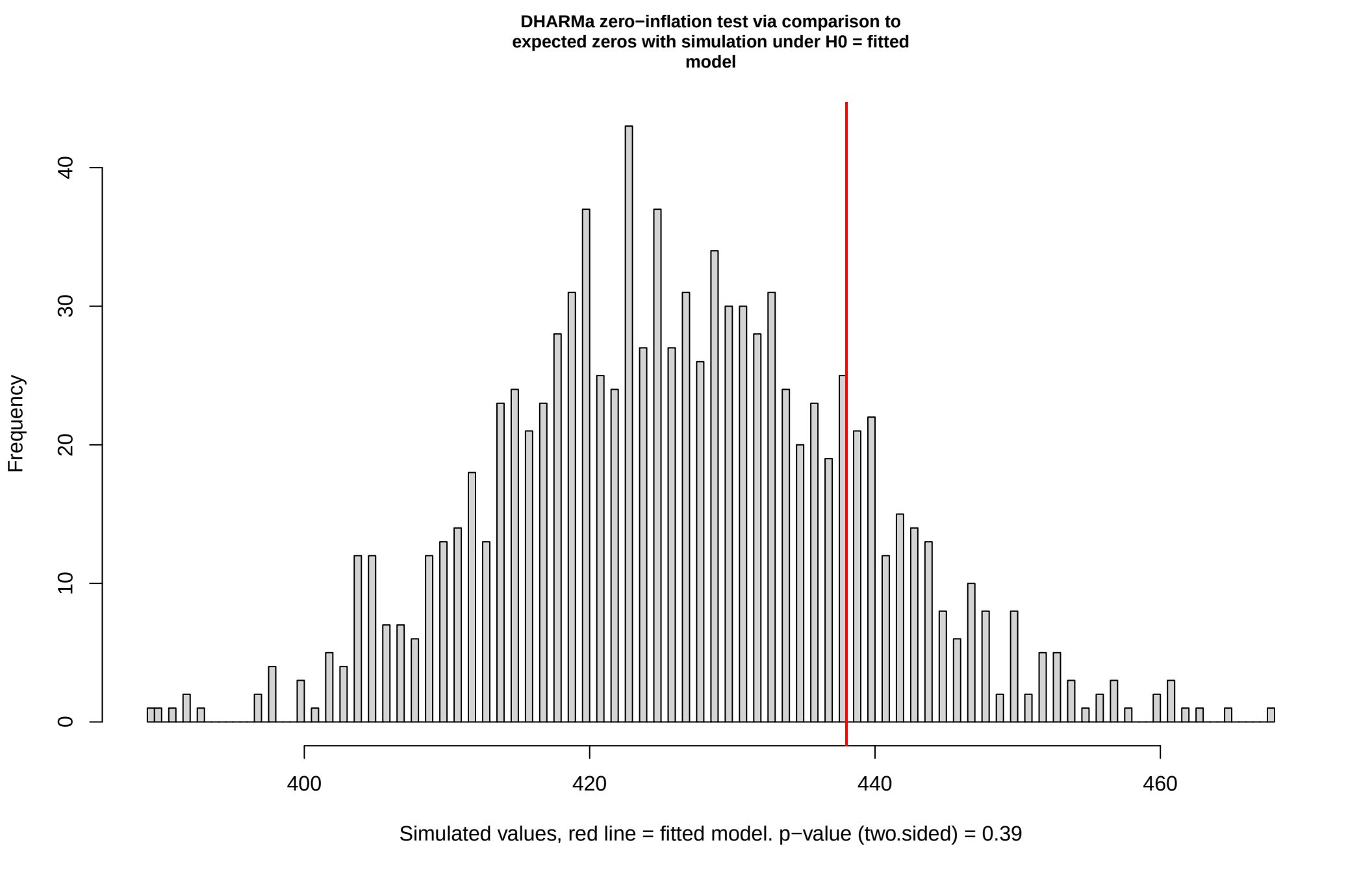

Residuals vs YearS
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

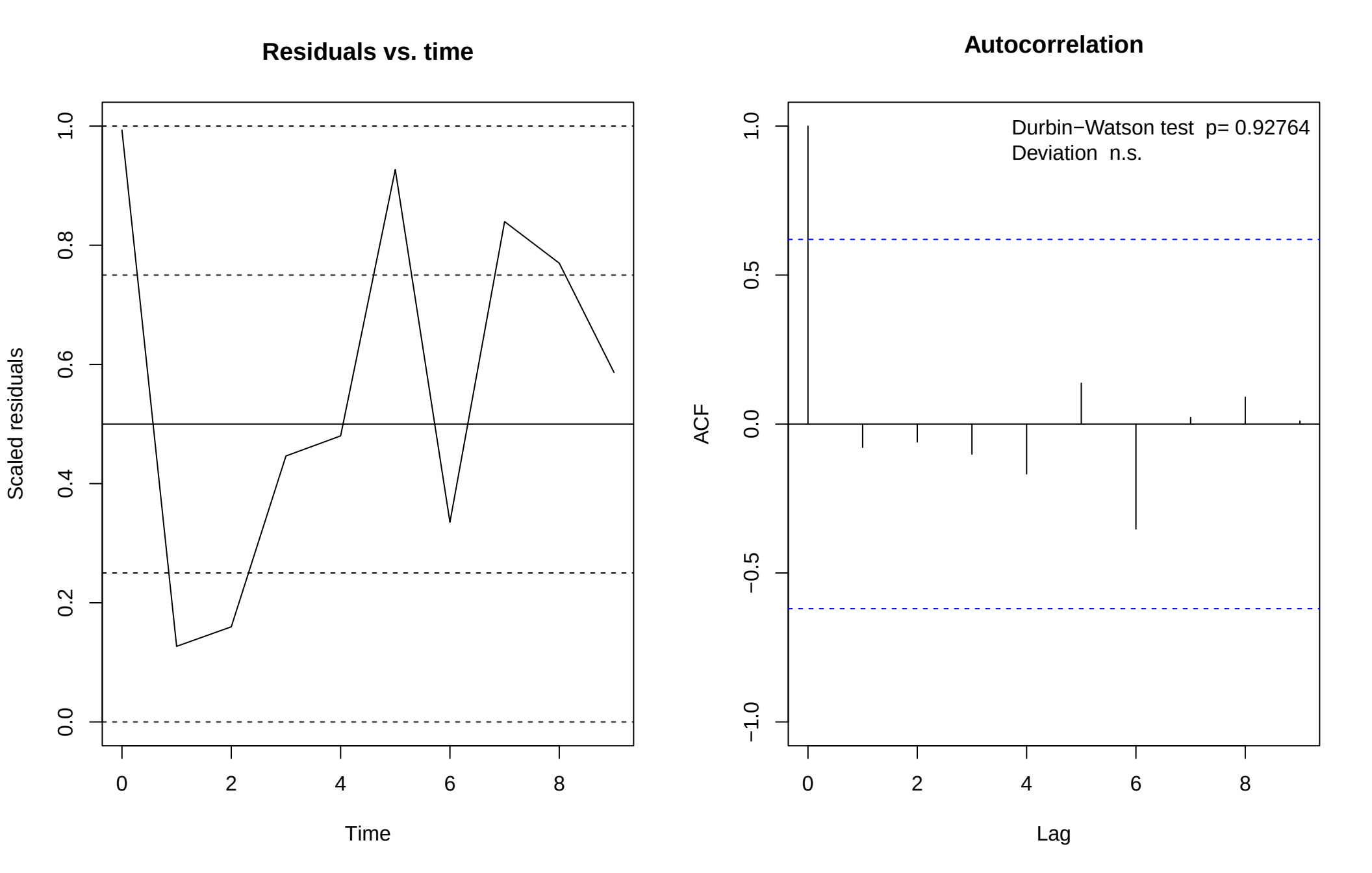

Residuals vs log(quad.nights)
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

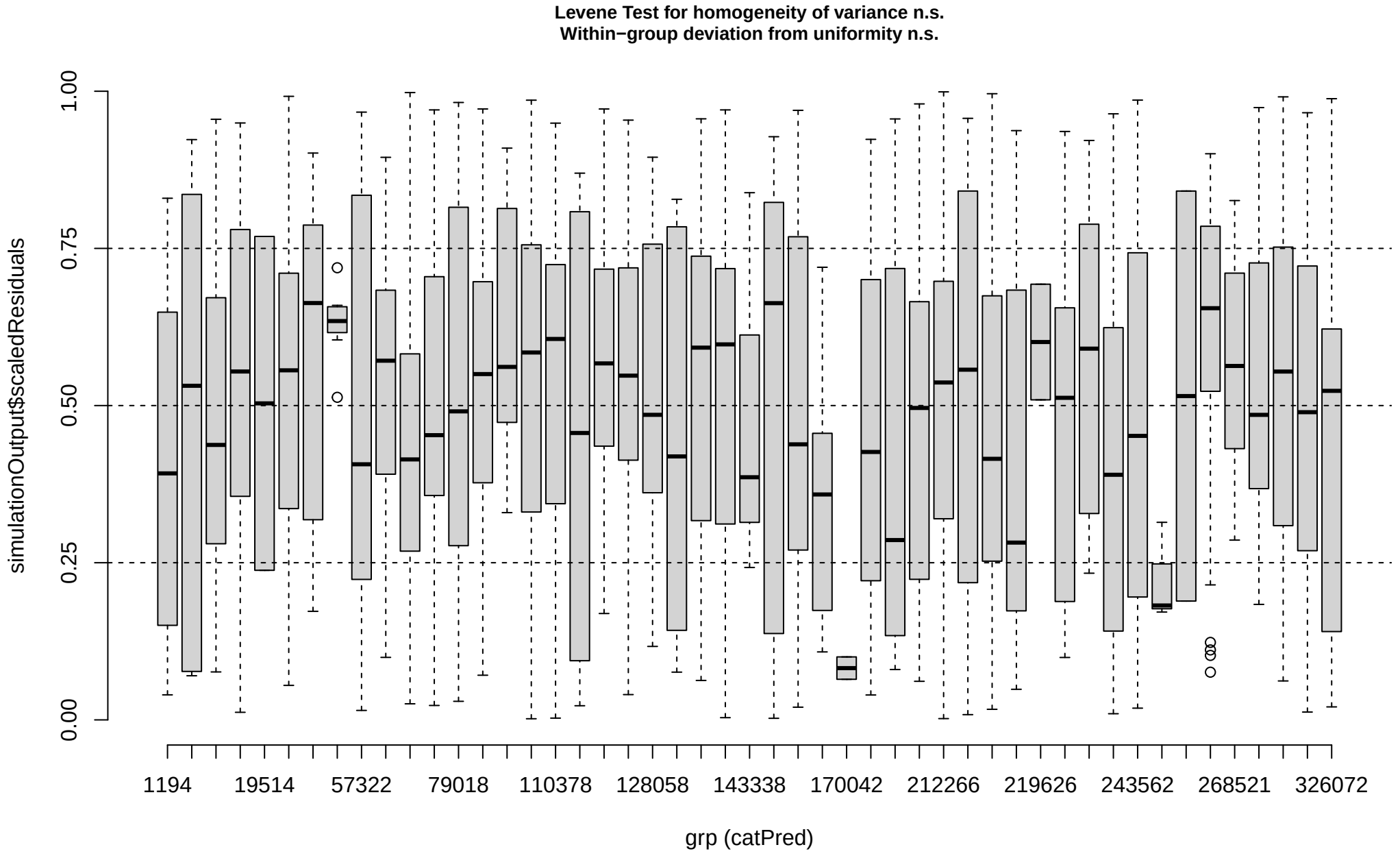

Residuals vs dd
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
dd (rank transformed)

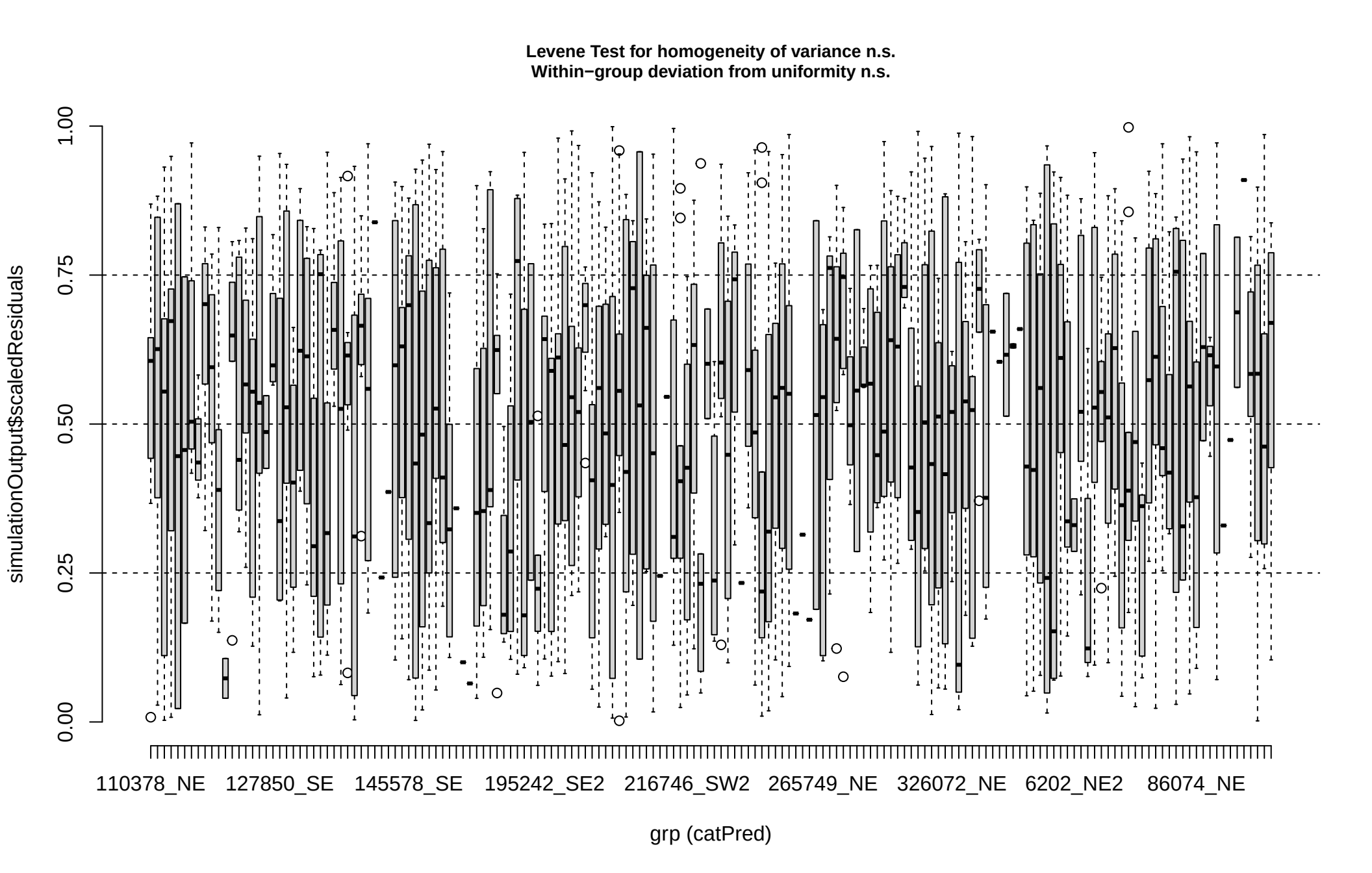

Residuals vs Nightly_Mean_Temp
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Temp (rank transformed)

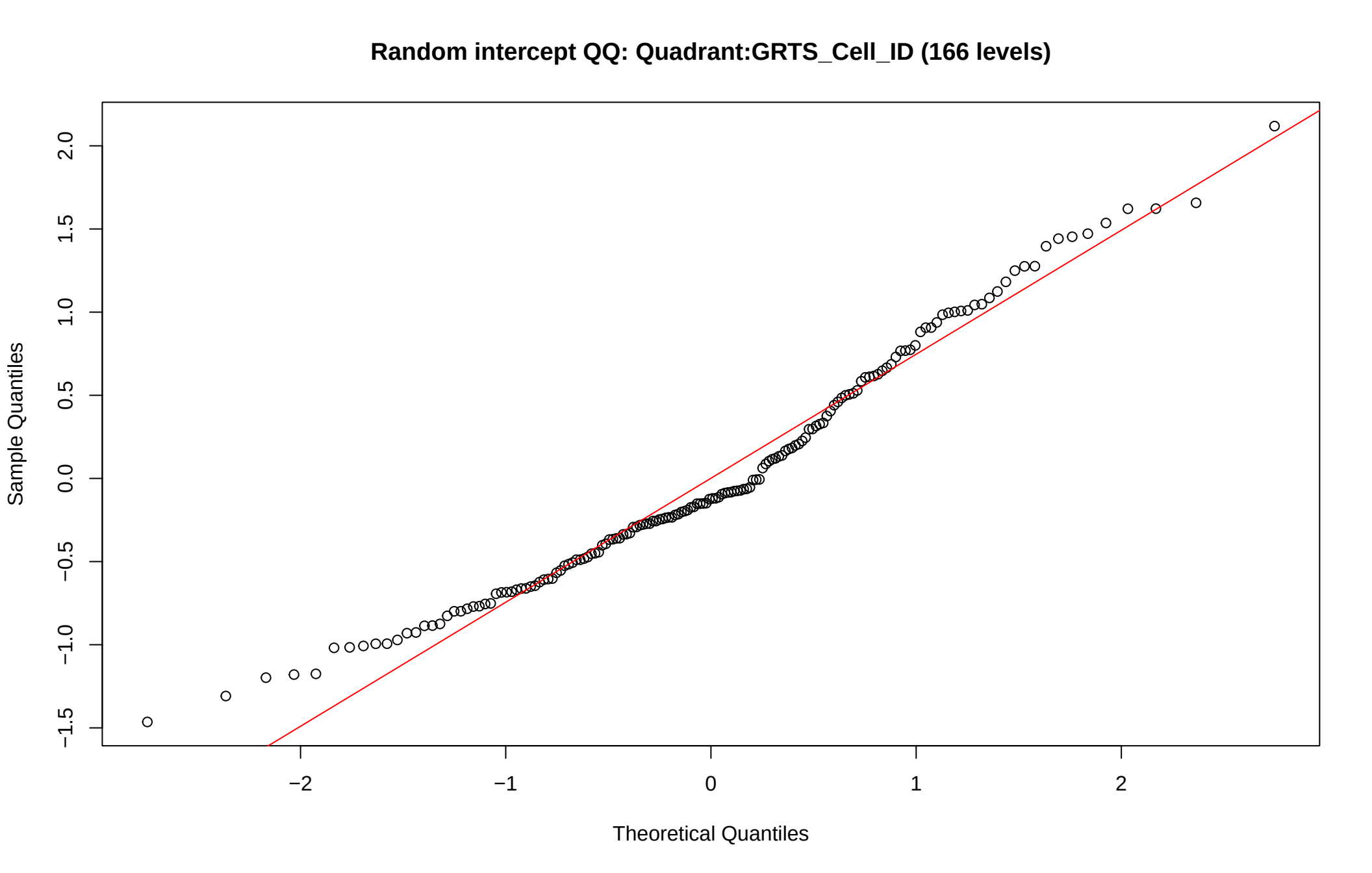

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
10
20
30
1
2
3
4
Simulated values, red line = fitted model. p−value (two.sided) = 0.9

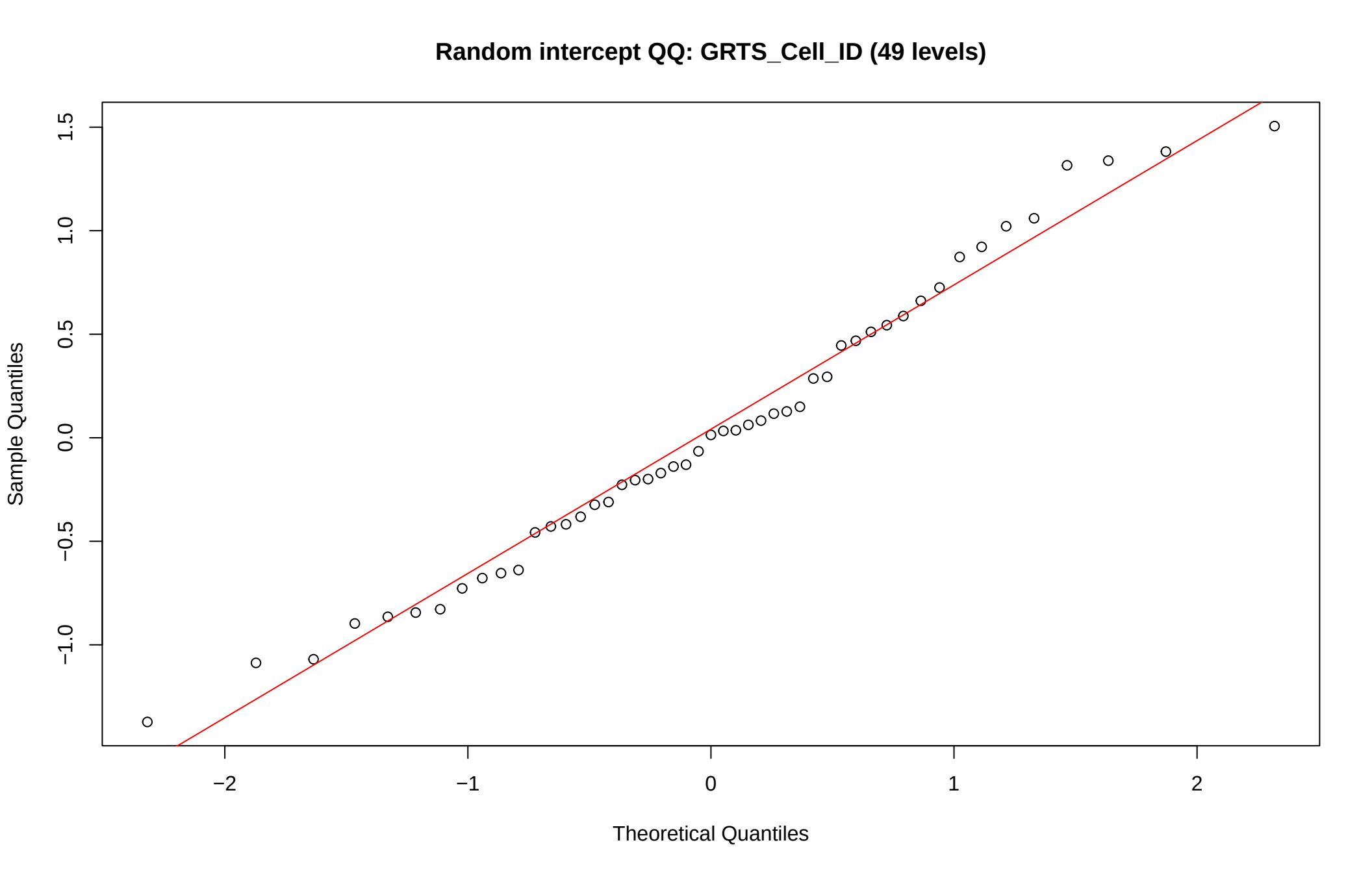

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
10
20
30
40
400
420
440
460
Simulated values, red line = fitted model. p−value (two.sided) = 0.39


Residuals vs. time
Scaled residuals
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Time
Autocorrelation
ACF
Durbin−Watson test p= 0.92764
Deviation n.s.
1.0
0.5
0.0
−0.5
−1.0
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
1194
19514
57322
79018
110378
128058
143338
170042
212266
219626
243562
268521
326072
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
110378_NE
127850_SE
145578_SE
195242_SE2
216746_SW2
265749_NE
326072_NE
6202_NE2
86074_NE
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (166 levels)
Sample Quantiles
Theoretical Quantiles
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2
−1
0
1
2


Random intercept QQ: GRTS_Cell_ID (49 levels)
Sample Quantiles
Theoretical Quantiles
1.5
1.0
0.5
0.0
−0.5
−1.0
−2
−1
0
1
2